Week 3 question sheet

Problem sheet

Instructions

Attempt all ten questions and all 17 scorable parts. The target time is 240 minutes and the total is 100 marks. No calculator or software is needed. Use exact values unless a comparison value is supplied. State supports, parameter restrictions and matrix or vector dimensions when they affect the method.

Use hints.md only after a serious attempt. Do not consult solutions.md until you have completed or paused every part.

Question 1 — membership in a span

Part A — `W03-Q01-PA` (5 marks; 14 minutes) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Determine whether $\mathbf{w} \in \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$. If it is, find the coefficients and verify the linear combination. If it is not, show the inconsistent coefficient condition.

Question 2 — independence, basis and coordinates

Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \mathbf{u}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \qquad \mathbf{u}_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Part A — `W03-Q02-PA` (5 marks; 14 minutes) Determine whether $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly independent. Give an explicit dependence relation and state the dimension of its span.

Part B — `W03-Q02-PB` (6 marks; 14 minutes) Show that the ordered set $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4)$ is a basis of $\mathbb{R}^3$. Then find the coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ for $\mathbf{x} = [3, 1, 2]^{\mathsf{T}}$. Explain why the coordinate vector is unique.

Question 3 — eigenvalues, eigenvectors and a diagonalisation

Let

$$\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Part A — `W03-Q03-PA` (5 marks; 14 minutes) Compute the characteristic polynomial and the eigenvalues of $\mathbf{A}$.

Part B — `W03-Q03-PB` (7 marks; 16 minutes) For each eigenvalue, find a corresponding non-zero eigenvector. Construct matrices $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. Verify the eigenvector relation in the compact form $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{D}$, and state why $\mathbf{P}$ is invertible.

Question 4 — diagonalisation and eigenbasis coordinates

Part A — `W03-Q04-PA` (10 marks; 20 minutes) Let

$$\mathbf{B} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}.$$

Diagonalise $\mathbf{B}$ over $\mathbb{R}$. Let

$$\mathbf{u} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

Write $\mathbf{u}$ in the ordered eigenbasis you use for the diagonalisation. Then use $\mathbf{B} = \mathbf{PDP}^{-1}$ to calculate $\mathbf{Bu}$, and verify the result by direct matrix multiplication. Your derivation must keep eigenvectors and eigenvalues in a consistent order.

Question 5 — a PMF, its CDF and moments

Let X have support $\{0, 1, 2, 3\}$ and PMF

$$p_X(x) = c(x + 1), \qquad x \in \{0, 1, 2, 3\}.$$

Part A — `W03-Q05-PA` (5 marks; 14 minutes) Find c, write the CDF $F_X(x)$ for every real x, and calculate $\mathbb{P}(1 < X \leq 3)$.

Part B — `W03-Q05-PB` (6 marks; 14 minutes) Derive $\mathbb{E}[X]$ and $\mathrm{Var}(X)$. Then let $Y = 3 - 2X$ and calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$, explaining why the additive constant affects one quantity but not the other.

Question 6 — required discrete distributions

Part A — `W03-Q06-PA` (5 marks; 14 minutes) Let $K \sim \mathrm{Binomial}(4, 1/2)$, where K counts successes in four independent Bernoulli trials, and let $G \sim \mathrm{Geometric}(1/4)$, where G counts trials up to and including the first success.

Calculate

$$\mathbb{P}(K = 2), \qquad \mathbb{P}(K \geq 1), \qquad \mathbb{P}(G > 3).$$

State the support of each random variable.

Part B — `W03-Q06-PB` (7 marks; 16 minutes) Let $N \sim \mathrm{Poisson}(2)$, where 2 is the mean/rate, and let K be as in Part A. Calculate $\mathbb{P}(N \leq 1)$. Then compare the means and variances of N and K. Explain how two count distributions can have the same mean but different variances.

Question 7 — required continuous distributions

Part A — `W03-Q07-PA` (5 marks; 14 minutes) Let $U \sim \mathrm{Uniform}(-1, 3)$. State its support and PDF, write its CDF for every real u, and calculate $\mathbb{P}(0 < U \leq 2)$.

Part B — `W03-Q07-PB` (7 marks; 16 minutes) Let $T \sim \mathrm{Exponential}(2)$, where 2 is the rate. State its support and PDF, calculate $\mathbb{P}(T > 1)$, and find its mean and variance. For $Y = 4T - 1$, calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$.

Question 8 — covariance, correlation and linear transformations

Part A — `W03-Q08-PA` (8 marks; 16 minutes) Random variables X and Y have finite second moments and satisfy

$$\mathbb{E}[X] = 1, \qquad \mathbb{E}[Y] = 2, \qquad \operatorname{Var}(X) = 4, \qquad \operatorname{Var}(Y) = 9, \qquad \operatorname{Cov}(X, Y) = -3.$$

Calculate $\operatorname{Corr}(X, Y)$. Then define

$$U = 2X - Y, \qquad V = X + Y.$$

Derive $\mathbb{E}[U]$, $\operatorname{Var}(U)$, $\operatorname{Var}(V)$ and $\operatorname{Cov}(U, V)$.

Question 9 — indicator variables and sampling without replacement

Part A — `W03-Q09-PA` (8 marks; 14 minutes) Two cards are drawn without replacement from a set of four cards, of which two are red. Let I_1 equal one if the first card is red and zero otherwise, and define I_2 analogously for the second card. Let $R = I_1 + I_2$.

Using the indicators, derive $\mathbb{E}[R]$, $\operatorname{Var}(R)$ and $\operatorname{Corr}(I_1, I_2)$. Interpret the sign of the correlation in the context of sampling without replacement.

Question 10 — earlier-outcome retrieval

Complete all three parts without consulting Week 1 or Week 2 notes.

Part A — `W03-Q10-PA` (`M4`; 3 marks; 10 minutes) Solve and classify the system

$$\begin{aligned} x + y + z &= 2, \\ 2x + 2y + 2z &= 4, \\ x - y &= 0. \end{aligned}$$

State the coefficient and augmented ranks and write the complete solution using a real free parameter.

Part B — `W03-Q10-PB` (`S2`; 4 marks; 10 minutes) Events A and A^c form a partition. Suppose

$$\mathbb{P}(A) = \frac{1}{3}, \qquad \mathbb{P}(B \mid A) = \frac{3}{4}, \qquad \mathbb{P}(B \mid A^c) = \frac{1}{4}.$$

Use total probability and Bayes' rule to calculate $\mathbb{P}(A \mid B)$. State the positivity condition needed for the posterior.

Part C — `W03-Q10-PC` (`M2`; 4 marks; 10 minutes) Throughout this part, log is the natural logarithm. Solve

$$\log(x - 1) - \log x = \log\left(\frac{1}{3}\right)$$

over the real numbers. State the domain before combining logarithms and check the final value in the original equation.

Self-test questions

Set aside 25 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. Show the condition, support or dimension check that justifies each conclusion. The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W03-ST01-PA` — eigenstructure and diagonalisation (`M5`; 5 marks; 7 minutes) Let

$$\mathbf{C} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

Find the eigenvalues and one eigenvector for each. Construct $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{C} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, and state why $\mathbf{P}$ is invertible.

`W03-ST02-PA` — a normal probability (`S3`; 5 marks; 6 minutes) Let $X \sim \mathcal{N}(10, 4)$, where the second parameter is the variance. State the support and standard deviation. Standardise X, then calculate

$$\mathbb{P}(8 \le X \le 12).$$

Use $\Phi(1) = 0.8413$ and symmetry of the standard normal CDF. Give the probability to four decimal places.

`W03-ST03-PA` — a transformed second moment (`S4`; 5 marks; 6 minutes) Suppose $\mathbb{E}[X] = 2$ and $\mathrm{Var}(X) = 3$. Let $Y = 5 - 2X$. Calculate $\mathbb{E}[Y]$, $\mathrm{Var}(Y)$ and $\mathbb{E}[Y^2]$.

`W03-ST04-PA` — a Poisson distribution (`S3`, `S4`; 5 marks; 6 minutes) Let $N \sim \mathrm{Poisson}(3)$, where 3 is the mean/rate. State the support, calculate $\mathbb{P}(N \le 1)$, and state $\mathbb{E}[N]$ and $\mathrm{Var}(N)$.